Personal Command Center

Full Life Operating System

Architecture & Cursor Build Plan — Version 2.2 — April 2026

Markets · Health · Spiritual Growth · Family · Productivity · Morning Routine

💹 Markets	💚 Health	🕋 Spiritual	👨‍👩‍👧 Family	☀️ Morning	🎙️ Media
Modules 01–11	Modules 12–15	Modules 16–19	Modules 20–22	Module 24	Module 23

1. Project Overview

Version 2.2 adds Module 24 — the Morning Routine Intelligence engine. Each morning, PCC reads last night's sleep data from Apple HealthKit, prompts you for a quick mood check-in (with your sleep score as the baseline), and uses that combined signal to recommend the optimal music/audio type, breakfast, and caffeine choice for your morning. Recommendations are explained, rateable, and improve over time. All prior content from v2.1 is fully preserved.

What's New in v2.2

- Module 24: Morning Routine Intelligence — sleep-aware, mood-aware daily recommendations
- Music rotation: Quran recitation, nasheeds, lo-fi, instrumental, classical/jazz, upbeat (pop/hip-hop), classic rock (80s & 90s)
- Caffeine logic: green tea (good sleep + calm), coffee 1 cup (moderate), coffee strong (poor sleep), with hydration reminders
- Breakfast logic: high-protein (eggs, Greek yogurt, protein shake) or traditional warm (oatmeal, toast, ful medames) based on energy need
- Delivery: Today dashboard card + full Morning Routine screen; no email
- Feedback loop: rate yesterday's recommendation (thumbs up/down + note), Claude learns your preferences over time

2. Today Dashboard (Morning Home Screen)

The Today dashboard is the app's home page. Prayer times remain the top card. The Morning Routine card appears second, immediately after prayer times.

Today Dashboard — Card Order (top to bottom)

🕌 **Prayer Times** Next prayer + countdown	☀️ **Morning Routine** Music, breakfast, caffeine	🕋 **Quran & Arabic** Today's goal, streak	💧 **Health** Calories, steps, sleep	📈 **Markets** Top signal, P&L	👨‍👩‍👧 **Family** Active project

- Morning Routine card shows: sleep score badge, mood selector (5 options), then recommendation summary (music genre, breakfast, caffeine) — all in one compact card
- Tap 'See full routine' to open the Morning Routine screen with full detail, reasoning, and audio playback
- Mood selector defaults to the AI-inferred mood from sleep data; you can tap to override in one tap
- On Fridays the Jumu'ah departure card replaces the Dhuhr entry as before

3. Module 24 — Morning Routine Intelligence

> *This is the primary new feature in v2.2. It combines Apple HealthKit sleep data, a lightweight mood check-in, a rule-based decision engine, Claude API narrative reasoning, and a feedback/learning loop to recommend music, breakfast, and caffeine each morning.*

3.1 Inputs — How Energy & Mood Are Determined

Step 1: Sleep baseline (automatic, from HealthKit)

- Total sleep duration pulled from the previous night's HealthKit sleep analysis record
- Sleep quality score computed from: duration, time-in-bed vs. asleep ratio, wake events, and deviation from 10 PM–6 AM target window
- Sleep tiers: Excellent (≥ 7.5 hrs, within 30 min of target) — Good (6–7.5 hrs) — Poor (< 6 hrs) — Overslept (missed Fajr window)
- The sleep tier becomes the energy baseline and is shown on the morning card as a colored badge (green / amber / red / gray)

Step 2: Mood check-in (optional 1-tap override)

- Mood selector appears on the Morning Routine card immediately after Fajr time
- 5 mood options displayed as icon + label: 😄 Energized — 😐 Neutral — 😪 Tired — 😰 Stressed — 🧘 Calm
- Default selection is AI-inferred from sleep tier: Excellent → Energized, Good → Neutral, Poor → Tired, Overslept → Stressed
- You tap to override if the default doesn't match how you actually feel — takes one tap, no typing
- Both the sleep tier and selected mood are stored and used together by the recommendation engine

3.2 Recommendation Engine — Decision Logic

The engine uses a two-layer approach: a fast rule-based layer for instant recommendations, and Claude for the explanatory narrative.

Layer 1: Rule-based recommendations (instant, zero API cost)

- Music/audio: mapped from mood × sleep tier using the decision matrix below
- Caffeine: green tea for good sleep + calm mood; standard coffee (1 cup) for moderate energy; strong coffee (1–2 cups) for poor sleep; always includes a hydration reminder if sleep was < 6 hrs
- Breakfast: high-protein options (eggs, Greek yogurt, protein shake) when energy is low or workout is scheduled; traditional warm options (oatmeal, toast, ful medames) when mood is calm or spiritual morning is planned; light options only when explicitly chosen or calorie budget is nearly met
- Calorie awareness: breakfast suggestion cross-references the health module's daily calorie goal and remaining budget
- Workout awareness: if a workout is scheduled in the exercise tracker for today, the engine always favors high-protein breakfast

Decision matrix: sleep × mood → recommendation

Sleep Score	Mood Override	Caffeine	Breakfast	Music / Audio
Excellent (≥ 7.5 hrs, on time)	Energized	Green tea (light)	High protein — eggs, Greek yogurt	Upbeat: classic rock 80s/90s, pop
Excellent (≥ 7.5 hrs, on time)	Calm	Green tea or none	Traditional warm — oatmeal, ful medames	Quran recitation or nasheed
Good (6–7.5 hrs)	Energized	Coffee (1 cup)	High protein — protein shake + toast	Classic rock or lo-fi
Good (6–7.5 hrs)	Tired / Stressed	Coffee (1 cup)	Traditional warm — oatmeal, toast	Quran recitation or instrumental
Poor (< 6 hrs)	Any	Coffee (strong, 1–2 cups)	High protein — eggs + protein shake	Lo-fi / ambient or nasheed
Poor (< 6 hrs)	Stressed	Coffee + water reminder	Light protein — Greek yogurt, smoothie	Quran recitation (calming)
Skipped Fajr (overslept)	Any	Coffee immediately	Quick high protein — shake or yogurt	Upbeat — classic rock or energetic nasheed

Layer 2: Claude narrative (one call per morning, ~$0.002)

- After the rule-based recommendation is generated, Claude receives: sleep duration, sleep quality score, mood selection, yesterday's rating, and today's calendar (prayer times, scheduled workout, any family activities)
- Claude returns a 2–3 sentence personalized morning message explaining the recommendation, e.g.: 'You got 5.5 hours last night and selected Tired — I've suggested a strong coffee and eggs to help you power through. Start with Quran recitation to center yourself before the day picks up. Your Asr prayer is at 4:30 PM — a good natural break point.'
- The narrative is displayed on the Morning Routine screen and on the Today card (collapsed to 1 sentence on the card)
- Cost: ~1 Claude call per morning × 30 days ≈ $0.06/month additional on top of existing Claude usage

3.3 Music & Audio Recommendation

Full rotation available (user-configured, all enabled by default)

- Quran recitation — Al Quran Cloud audio (same source as Spiritual pillar). Recommended for: calm mood, post-Fajr mornings, stressed state (grounding)
- Arabic nasheeds (no instruments) — curated playlist links (YouTube or local files). Recommended for: spiritual mornings, calm or neutral mood
- Lo-fi / ambient — YouTube playlist links (curated lofi channels). Recommended for: tired state, focus-needed mornings, poor sleep recovery
- Instrumental / nature sounds — YouTube or Spotify-style links. Recommended for: calm + good sleep, meditation mornings
- Classical / jazz — YouTube playlist links. Recommended for: good sleep + neutral mood, work-from-home mornings
- Upbeat / energetic (pop, hip-hop) — YouTube playlist links. Recommended for: excellent sleep + energized mood, workout mornings
- Classic rock — 80s & 90s (e.g. Bon Jovi, Queen, Eagles, Def Leppard). YouTube playlist links. Recommended for: good/excellent sleep + energized mood, overslept recovery boost

Music delivery

- Morning Routine screen shows the recommended genre with a 'Play' button that opens a curated YouTube playlist in a new tab or the in-app YouTube embed
- You can tap any other genre tile to switch — all 7 options are shown as tappable cards
- Selected genre is logged for feedback tracking

> ⚠ *Quran recitation is always available regardless of recommendation — a dedicated 'Listen to Quran' shortcut appears at the top of the music section.*

3.4 Caffeine Recommendation

- Green tea: recommended when sleep was Excellent + mood is Calm or Neutral. Displays: 'Green tea — you're well-rested, light caffeine is enough.'
- Coffee — 1 cup: recommended when sleep was Good or mood is Tired/Energized with Good sleep. Displays brew suggestion (e.g. drip, pour-over).
- Coffee — strong (1–2 cups): recommended when sleep was Poor (< 6 hrs). Displays: 'Strong coffee — you need the boost today. Drink a glass of water first.'
- Hydration reminder: always shown when sleep < 6 hrs or mood is Stressed — 'Drink 16 oz water before your coffee.'
- Skip caffeine option: always available with one tap — no judgment
- Logged to the food_log table as a zero-calorie entry for tracking consistency

3.5 Breakfast Recommendation

High-protein options (recommended for low energy, post-workout, poor sleep)

- Eggs — scrambled, fried, or boiled. Shown with estimated prep time (5–8 min) and calorie count (~200–300 kcal depending on count)
- Greek yogurt — with honey or fruit. Shown with calorie count (~150–200 kcal)

- Protein shake — shown with suggested flavor/brand if user has configured a preference. (~200–350 kcal)
- Eggs + protein shake combo: suggested when sleep was Poor and a workout is scheduled today

Traditional / warm options (recommended for calm mornings, spiritual focus, good sleep)

- Oatmeal — with dates, honey, or banana. Calorie estimate shown (~300–400 kcal). Prep time: 5 min.
- Toast — whole grain with peanut butter or avocado. Calorie estimate (~250–350 kcal)
- Ful medames — Egyptian fava bean dish. Shown as 'Weekend option' (longer prep). Calorie estimate (~350–450 kcal)

Breakfast calorie integration

- Suggested breakfast calorie count is automatically pre-filled into the food log as a pending entry — one tap to confirm and log it
- If the remaining daily calorie budget is less than the suggested breakfast, the engine substitutes a lighter option and notes the reason
- Breakfast skip option always available — skipping is logged and factored into calorie budget for the day

3.6 Morning Routine Screen

A dedicated full-screen view accessible from the Today dashboard card. Opens after Fajr. Contains:

- Sleep summary banner: last night's duration, quality score, sleep tier badge, deviation from 10 PM–6 AM target
- Mood selector: 5-option one-tap picker, defaulted to AI inference, overridable
- Recommended music genre card: genre name, description, Play button (opens YouTube playlist), all 7 genres shown as tappable alternates below
- Caffeine card: recommendation with brief reasoning, one-tap 'Done' to log it, 'Skip' option
- Breakfast card: recommended option with calorie estimate, prep time, one-tap log-to-food-diary, 'Skip' option
- Claude narrative: 2–3 sentence personalized morning message
- Yesterday's rating: thumbs up / thumbs down + optional 1-line note. Appears at the bottom as a gentle prompt ('How was yesterday's morning routine?')

3.7 Feedback & Learning Loop

- Each morning's recommendation (music genre, caffeine, breakfast) is stored in the morning_log table with the sleep score and mood at the time
- Each evening (or next morning), a rating prompt appears: thumbs up (kept the recommendation) or thumbs down (overrode it), plus an optional note
- After 14 days of ratings, Claude analyzes the pattern: 'You consistently override the lo-fi recommendation when your sleep is Good — switching your Good sleep default to Classic Rock.'

- Preference adjustments are stored in morning_preferences table and override the default decision matrix going forward
- The learning loop runs as a weekly background job (APScheduler) — one Claude call per week analyzing the past 7 days of ratings (~$0.005/week)

3.8 Morning Routine Database Schema

- morning_log: id, date, sleep_duration_min, sleep_quality_score, sleep_tier, mood_inferred, mood_selected, music_genre_recommended, music_genre_used, caffeine_recommended, caffeine_used, breakfast_recommended, breakfast_used, claude_narrative, created_at
- morning_ratings: id, morning_log_id, rating (up/down), note, rated_at
- morning_preferences: id, sleep_tier, mood, music_genre_override, caffeine_override, breakfast_override, confidence_score, last_updated
- morning_music_playlists: id, genre, label, youtube_url, description, active

3.9 Morning Routine Backend Endpoints

- GET /api/morning/recommendation — returns today's full recommendation (sleep data + mood default + all three suggestions + Claude narrative)
- POST /api/morning/mood — saves mood override {mood: 'tired'}
- POST /api/morning/log — saves what was actually used {music_genre_used, caffeine_used, breakfast_used}
- POST /api/morning/rating — saves rating {morning_log_id, rating, note}
- GET /api/morning/history?days=30 — returns 30-day morning log with ratings
- POST /api/morning/preferences/recalculate — triggers the weekly Claude preference analysis job

3.10 Cursor Prompt for Module 24

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Build the Morning Routine Intelligence module for PCC. Stack: FastAPI + PostgreSQL +
Redis + Claude API. Inputs: (1) last night's HealthKit sleep data from the
health_metrics TimescaleDB table, (2) mood selected by user (fajr/dhuhr/asr/maghrib/isha
enum). Sleep tiers: Excellent >=7.5hrs, Good 6-7.5hrs, Poor <6hrs. Decision matrix: Poor
sleep -> strong coffee + high protein breakfast + lofi/quran. Good sleep + tired mood ->
1 cup coffee + warm breakfast + lofi. Good sleep + energized -> green tea or 1 cup
coffee + high protein + classic rock or upbeat. Excellent sleep + calm -> green tea +
traditional warm + quran or nasheed. Music genres stored in morning_music_playlists
table. After rule-based recommendation, call Claude API with: sleep_duration,
sleep_tier, mood, yesterday_rating, today_prayer_times, today_workout_scheduled. Claude
returns 2-3 sentence narrative. Weekly APScheduler job calls Claude to analyze 7 days of
ratings and update morning_preferences table. Endpoints: GET
/api/morning/recommendation, POST /api/morning/mood, POST /api/morning/log, POST
/api/morning/rating, GET /api/morning/history. Write all 4 DB tables and pytest tests
for all endpoints.
```

4. Pillar A — Markets & Productivity

Unchanged from v2.1. Modules 01–11 cover market data, macro indicators, news aggregation, portfolio management, the signal engine, Gmail/Outlook/Teams/TDx inbox digest, Google Keep, Charles Schwab sync, markets frontend, and Docker/Caddy deployment.

Markets Pillar — Key Screens					
Market Dashboard Quotes, VIX, heatmap	**Portfolio Manager** P&L, targets, margins	**News Feed** Energy, metals, sectors	**Macro Strip** CPI, PCE, employment	**Signal Config** Weights + narrative	**Inbox Digest** Gmail, Outlook, TDx

Modules 01–11: scaffold, market data, macro (FRED), news, portfolio, signal engine (rule-based scorer + Claude narrative), inbox digest (Gmail OAuth + Graph API + TeamDynamix REST), Google Keep, Charles Schwab OAuth sync, markets frontend (Next.js + Recharts), Docker Compose + Caddy HTTPS deployment.

5. Pillar B — Physical Health

Unchanged from v2.1. Modules 12–15.

Module 12 — iOS HealthKit Companion

- SwiftUI app: reads HealthKit (calories, steps, swim, walk, HR, weight, sleep analysis), POSTs to PCC backend every 15 min (foreground) / hourly (background)
- Also delivers prayer reminder APNs push notifications (time-sensitive, breaks through iOS Focus)
- Also delivers Morning Routine push at wake time (6:00 AM) with sleep score badge and mood selector deep-link

Module 13 — Health Data Service

- HealthKit ingest endpoint, TimescaleDB hypertable for health_metrics, daily rollup table
- USDA FoodData Central API for calorie logging food search
- Calorie budget and TDEE computation; macros ring chart data

Module 14 — Exercise Tracker

- Weekly targets per type (swimming, walking, weight training), AI weekly plan via Claude API
- HealthKit auto-import for cardio; manual log for resistance training
- Cardio health: resting HR trend, VO2 max estimate, heart rate zones
- Today's workout schedule exposed to Morning Routine engine (Module 24) for breakfast logic

Module 15 — Health Frontend

- Today health card: calorie ring, steps, sleep score badge, active calories
- Sleep chart: 30-day rolling average, sleep score trend, deviation from 10 PM/6 AM target
- Exercise rings: weekly completion per type, HR zone chart

- Weight chart: actual vs. goal trajectory

6. Pillar C — Spiritual Development

Unchanged from v2.1. Modules 16–19.

Module 16 — Prayer, Adhan & Salah Tracker

- Prayer times: adhan-py library, ISNA method, Sterling VA coordinates (38.9898° N, 77.3897° W)
- Adhan audio: Mishary Rashid Al-Afasy from cdn.islamic.network. Fajr version includes al-salatu khayrun min al-nawm.
- Tri-channel reminders: iPhone APNs push 20 min before + in-app WebSocket banner + adhan plays at prayer time
- Salah log: On Time / Qada / Missed × Masjid / Home / Work / Traveling, one-tap from Today dashboard
- Jumu'ah: Dhuhr replaced on Fridays. Adams Center (46903 Sugarland Rd, Sterling VA). Imam Majed / Imam Ouertani. Departure reminder (default 60 min early). Khutba notes.
- History: 5-segment daily prayer ring (green/amber/red/gray), 7-ring weekly row, monthly stats, location pie chart

Prayer	Arabic	Typical Time*	Adhan	20-min Alert	Reminder	Track Status
Fajr	فجر	5:15 AM	✅	✅	iPhone push	On time / Qada / Missed
Dhuhr	ظهر	1:00 PM	✅	✅	Push + banner	On time / Qada / Missed
Asr	عصر	4:30 PM	✅	✅	Push + banner	On time / Qada / Missed
Maghrib	مغرب	8:10 PM	✅	✅	Push + banner	On time / Qada / Missed
Isha	عشاء	9:40 PM	✅	✅	Push + banner	On time / Qada / Missed

** Approximate for Sterling, VA in April. Calculated daily by adhan-py using ISNA method.*

Module 17 — Quran & Arabic Backend

- Al Quran Cloud API + Quran.com API: verse text, word-level data, translations, audio
- SM-2 spaced repetition for hifz scheduling; streak logic across prayer + Quran + Arabic activity

Module 18 — Arabic Learning UI

- Phase 1: 28-letter guide with forms, audio (Web Speech API), canvas tracing
- Phase 2: verse-by-verse reader with transliteration and tap-to-hear audio

Module 19 — Hifz Tracker UI

- Flashcard quiz with SM-2 queue, progress bars (X/28 letters, X/6,236 ayahs), streak display

7. Pillar D — Family

Unchanged from v2.1. Modules 20–22.

Module 20 — Family Data Service

- Activity DB, project/milestone/task/dependency schema, Claude API suggestion endpoint
- Family members: daughter (10, interests: STEM, Art, Soccer) + wife (interests: business, side hustle, personal development)

Module 21 — Activity Planner UI

- AI suggests 3 activities/week per family member based on age + interests; you approve or skip
- Kanban activity board: Planned / In Progress / Done. Photo upload per completed activity.

Module 22 — Side Hustle PM UI

- Full PM: projects, milestones, tasks, dependencies. Board view (Trello-style) + Timeline view (Gantt).
- Weekly Claude project health summary ('You have 3 overdue tasks and 2 milestones at risk')

8. Pillar E — Media Queue

Unchanged from v2.1. Module 23.

Module 23 — Media Queue Service

- YouTube Data API v3: subscription-based, interest-tagged, 30-min daily playlist via Claude curation
- Podcast RSS + iTunes Search API: episode library, in-app player (speed control, 15s skip, sleep timer)
- Apple Podcasts handoff via podcast:// URL deep link
- Unified queue view rebuilt each morning alongside Today dashboard refresh

9. Complete Module List — v2.2 (24 Modules)

#	Pillar	Module	Description	Key Tech
01	**Markets**	Project Scaffold	Monorepo, Docker Compose, .env, Caddy config	Docker, Caddy, Python, Node
02	**Markets**	Market Data Service	Yahoo Finance + Alpha Vantage + Finnhub. Redis cache. TimescaleDB OHLCV.	yfinance, FastAPI, Redis
03	**Markets**	Macro Data Service	FRED: CPI, PCE, unemployment, payrolls. 6hr cache.	FRED API, FastAPI
04	**Markets**	News Aggregation	NewsAPI by topic + symbol. URL dedup. Keyword sentiment.	NewsAPI, Redis SET
05	**Markets**	Portfolio Manager	CRUD positions. Live P&L. Target sell price + margin.	PostgreSQL, FastAPI
06	**Markets**	Signal Engine	Plugin scorer + weight store + hourly Claude narrative.	Claude API, PostgreSQL
07	**Markets**	Unified Inbox Digest	Gmail OAuth + Graph API + TDx tickets. AI priority summary.	Google API, Graph API
08	**Markets**	Google Keep Integration	gkeepapi reader. Date regex. Upcoming reminders endpoint.	gkeepapi, Python
09	**Markets**	Schwab Sync	OAuth 2.0 position sync to portfolio table. Daily P&L.	Schwab API, OAuth 2
10	**Markets**	Markets Frontend	Market, Portfolio, News, Deep Dive, Signal Config pages. Responsive.	Next.js, Tailwind, Recharts
11	**Markets**	Docker + Deployment	Final compose.yml, Caddyfile HTTPS, deploy.sh zero-downtime.	Docker Compose, Caddy
12	**Health**	iOS HealthKit Companion	SwiftUI: HealthKit read, POST to backend, APNs prayer + morning push notifications.	SwiftUI, HealthKit, APNs
13	**Health**	Health Data Service	HealthKit ingest, TimescaleDB, calorie/macro endpoints, USDA food search.	FastAPI, TimescaleDB, USDA
14	**Health**	Exercise Tracker	Weekly targets, workout log, AI plan, VO2/HR zones. Exposes today's workout to Module 24.	FastAPI, Claude API
15	**Health**	Health Frontend	Today health card, calorie ring, sleep chart, exercise rings, weight chart.	Next.js, Recharts
16	**Spirit.**	Prayer, Adhan & Salah	adhan-py ISNA calc, Mishary Al-Afasy audio, tri-channel reminders, salah log, Jumu'ah, rings.	adhan-py, FastAPI, APNs
17	**Spirit.**	Quran & Arabic Backend	Al Quran Cloud + Quran.com APIs. SM-2 spaced repetition. Streaks.	FastAPI, PostgreSQL, Redis
18	**Spirit.**	Arabic Learning UI	Letter guide, canvas tracing, pronunciation audio, verse reader.	Next.js, Canvas API, Web Speech
19	**Spirit.**	Hifz Tracker UI	Flashcard quiz, spaced repetition queue, progress bars, streak display.	Next.js, shadcn/ui
20	**Family**	Family Data Service	Activity DB, project/milestone/task schema. Claude AI suggestion endpoint.	FastAPI, PostgreSQL, Claude API
21	**Family**	Activity Planner UI	Daughter + wife boards. AI suggestion inbox. Kanban completion + photo log.	Next.js, DnD, shadcn/ui
22	**Family**	Side Hustle PM UI	Project board, Gantt milestones, task dependencies, weekly AI health summary.	Next.js, Recharts, Claude API

23	**Media**	Media Queue Service	YouTube Data API + Podcast RSS. 30-min queue builder. In-app audio player.	YouTube API, iTunes API, FastAPI
24	**Morning**	Morning Routine Intelligence	Sleep-aware + mood-aware daily music/caffeine/breakfast recommendations. Claude narrative. Feedback learning loop.	FastAPI, Claude API, adhan-py

10. Complete Database Schema (v2.2)

Markets

- positions: id, symbol, shares, purchase_price, purchase_date, target_sell_price, profit_margin_pct, notes
- transactions: id, symbol, type, shares, price, date
- signal_weights: id, plugin_name, weight, updated_at
- signal_history (TimescaleDB): time, symbol, score, breakdown_json, narrative
- news_articles: id, url_hash, title, source, published_at, topics_json, sentiment_score

Health

- health_metrics (TimescaleDB hypertable): time, metric_type, value, source
- daily_health_rollup: date, steps, active_cal, resting_cal, swim_min, walk_min, weight_kg, sleep_min, sleep_quality
- food_log: id, date, food_name, calories, protein_g, carbs_g, fat_g, created_at
- exercise_log: id, date, type, duration_min, sets_json, notes
- health_goals: metric, target_value, target_date, created_at

Prayer

- prayer_times: id, date, fajr, dhuhr, asr, maghrib, isha, calculation_method, latitude, longitude, created_at
- prayer_log: id, date, prayer, status (on_time/qada/missed), location (masjid/home/work/travel), notes, logged_at
- jumuah_log: id, date, location, imam, confirmed_time, khutba_notes, logged_at
- prayer_settings: id, calculation_method, asr_method, latitude, longitude, adhan_volume, muted_prayers_json, leave_early_minutes

Spiritual

- arabic_letters: id, letter, name, forms_json, audio_url, mastered_at
- letter_practice_log: id, letter_id, date, score, duration_sec
- quran_progress: id, surah, ayah, status, next_review_date, interval_days, ease_factor
- spiritual_streaks: date, activity_type, minutes_logged

Morning Routine (NEW in v2.2)

- morning_log: id, date, sleep_duration_min, sleep_quality_score, sleep_tier, mood_inferred, mood_selected, music_genre_recommended, music_genre_used, caffeine_recommended, caffeine_used, breakfast_recommended, breakfast_used, claude_narrative, created_at
- morning_ratings: id, morning_log_id, rating (up/down), note, rated_at
- morning_preferences: id, sleep_tier, mood, music_genre_override, caffeine_override, breakfast_override, confidence_score, last_updated
- morning_music_playlists: id, genre, label, youtube_url, description, active

Family

- family_members: id, name, relationship, interests_json, birth_date
- activity_ideas: id, member_id, title, description, category, materials, duration_min, ai_generated, approved, created_at
- activity_log: id, idea_id, completed_date, notes, photo_url
- projects: id, name, description, goal, target_date, status
- milestones: id, project_id, title, target_date, completed_at
- tasks: id, milestone_id, title, owner, due_date, status, priority, blocking_task_ids_json

Media

- podcast_feeds: id, title, rss_url, interest_tags_json, last_fetched
- podcast_episodes: id, feed_id, title, audio_url, duration_sec, published_at, playback_position_sec, completed
- youtube_channels: id, channel_id, title, interest_tags_json
- media_queue: id, date, item_type, item_id, order, completed

11. Complete Data Source Registry

Source	Cost	Data	Limit	Use Case
Yahoo Finance (yfinance)	Free	Quotes, OHLCV	Unlimited	Primary market data
Alpha Vantage	Free	RSI, MACD, OHLCV	25/day	Technical indicators
FRED API	Free	CPI, PCE, jobs	Unlimited	Macro data
NewsAPI	Free	Global news	100/day	Market & sector news
Finnhub	Free	VIX, earnings	60/min	Volatility data
Polygon.io	~$29	Options, futures	Unlimited	Options deep dive
Apple HealthKit	Free	Activity, sleep, HR, weight	Device-local	All health + morning sleep input
USDA FoodData Central	Free	600k+ foods with macros	3,600/hr	Calorie logging

adhan-py (local library)	Free	Prayer time calc (ISNA)	Local, no API	Prayer times for all 5 prayers
cdn.islamic.network	Free	Mishary Al-Afasy adhan MP3s	CDN	Adhan playback for 5 prayers
Al Quran Cloud API	Free	Quran text + audio	Unlimited	Quran recitation + morning audio
Quran.com API	Free	Word-level data	Generous	Word-by-word reading
YouTube Data API v3	Free	Subscriptions, videos	10k units/day	Media queue + music playlists
iTunes Search API	Free	Podcast directory	Unlimited	Podcast search
Claude API (Sonnet 4)	~$3/mo	LLM inference	Per token	Signal narrative, activity suggestions, morning routine reasoning, learning loop

12. Updated Monthly Cost Estimate (v2.2)

Item	Cost/mo	Notes
Yahoo Finance / FRED / Finnhub / Quran / iTunes / adhan audio / adhan-py	Free	Core data — zero cost
Alpha Vantage	Free	25 calls/day covers indicators
NewsAPI	Free	100 calls/day sufficient
USDA FoodData Central	Free	3,600 req/hr — far more than needed
YouTube Data API v3	Free	10k units/day covers subscription reads + music playlists
Claude API — market signal narrative (hourly)	~$1.50/mo	~1,000 calls/mo at Sonnet pricing
Claude API — morning routine (1 call/morning + weekly learning)	~$0.10/mo	30 morning calls + 4 weekly analysis calls
Claude API — activity suggestions + project health	~$1.40/mo	Family + side hustle weekly summaries
Claude API — total	~$3/mo	All pillars combined
Polygon.io (optional — options data)	~$29/mo	Only if real-time options chain needed
VPS hosting (Phase 2, optional)	$0–$5/mo	GCP free tier or Hetzner CX11
TOTAL — Phase 1 baseline (no options, no VPS)	**~$3/mo**	**All 6 pillars + 24 modules running**
TOTAL — Phase 2 full (options data + VPS)	**~$37/mo**	**Full feature set, self-hosted HTTPS**

13. Recommended Build Order for Cursor

Group 1 — Foundation

1. Module 01: Scaffold — Docker Compose, folder structure, .env
2. Module 02 + 03: Market data + Macro data
3. Module 13: Health data service + TimescaleDB schema (morning routine reads from this)

Group 2 — Core backends (parallelize freely)

4. Module 05: Portfolio Manager
5. Module 04: News Aggregation
6. Module 16: Prayer, Adhan & Salah — build early so prayer times feed Today dashboard
7. Module 17: Quran & Arabic backend
8. Module 20: Family data service
9. Module 23: Media queue service
10. Module 24: Morning Routine Intelligence — depends on Module 13 (sleep data) and Module 14 (workout schedule)

Group 3 — Intelligence layer

11. Module 06: Signal Engine (depends on 02, 03, 04)
12. Module 14: Exercise Tracker + AI weekly plan (depends on 13; exposes workout schedule to Module 24)

Group 4 — Integrations

13. Module 07: Gmail + Outlook + TDx digest
14. Module 08: Google Keep
15. Module 09: Charles Schwab sync
16. Module 12: iOS companion app (SwiftUI — separate Xcode project; delivers HealthKit data, prayer APNs, and 6 AM morning push)

Group 5 — Frontend (build last)

17. Module 10: Markets frontend
18. Module 15: Health frontend
19. Module 18 + 19: Arabic learning + Hifz tracker UI
20. Module 21 + 22: Family activity planner + Side hustle PM
21. Morning Routine screen UI (part of Module 24): sleep banner, mood picker, music/caffeine/breakfast cards, Claude narrative, rating prompt
22. Today Dashboard: build last — aggregates prayer times, morning routine card, and all pillar summaries

Group 6 — Deployment

23. Module 11: Final Docker Compose, Caddy, deploy.sh zero-downtime script

14. Cursor Session Templates

Standard session header (all modules)

```
Stack: Next.js 14 / FastAPI / PostgreSQL+TimescaleDB / Redis / Docker Compose. This is a
personal life OS called PCC with 6 pillars: Markets, Health, Spiritual, Family, Media,
Morning Routine. We are on Module [N]: [Name]. Here is the DB schema: [paste]. Here are
prior API contracts: [paste]. Build only this module. Write pytest tests for all
endpoints. Do not modify other modules.
```

Module 16 — Prayer Cursor prompt

```
Build the Prayer, Adhan & Salah module. Use adhan-py (pip install adhan) with ISNA
method, coordinates 38.9898,-77.3897 (Sterling VA). Adhan: Mishary Al-Afasy from
cdn.islamic.network; Fajr version includes al-salatu khayrun min al-nawm. APScheduler:
schedule WebSocket pushes 20 min before each prayer and at prayer time. POST
/api/prayer/log: {prayer, status (on_time/qada/missed), location
(masjid/home/work/travel), notes}. Fridays: replace dhuhr with jumuah (Adams Center,
46903 Sugarland Rd Sterling VA, imams: Mohammed Majed / Abd Ar-Rafa Ouertani,
configurable leave_early_minutes default 60). 4 DB tables: prayer_times, prayer_log,
jumuah_log, prayer_settings.
```

Module 24 — Morning Routine Cursor prompt

```
Build the Morning Routine Intelligence module. Inputs: (1) prior night HealthKit sleep
from health_metrics TimescaleDB table, (2) user mood
(energized/neutral/tired/stressed/calm). Sleep tiers: Excellent >=7.5hrs, Good 6-7.5hrs,
Poor <6hrs. Decision matrix: Poor -> strong coffee + high protein + lofi/quran.
Good+tired -> 1 cup coffee + warm breakfast + lofi. Good+energized -> green tea or
coffee + protein + classic rock or upbeat. Excellent+calm -> green tea + warm
traditional + quran or nasheed. After rule engine, call Claude API (sleep_duration,
sleep_tier, mood, yesterday_rating, today_prayer_times, today_workout_scheduled) for 2-3
sentence narrative. Weekly APScheduler job: Claude analyzes 7 days of ratings, updates
morning_preferences table. Endpoints: GET /api/morning/recommendation, POST
/api/morning/mood, POST /api/morning/log, POST /api/morning/rating, GET
/api/morning/history. 4 DB tables: morning_log, morning_ratings, morning_preferences,
morning_music_playlists.
```

Module 12 iOS note: open a separate Xcode project. App reads HealthKit and POSTs to http://[local-ip]:8000/api/health/ingest. Register for APNs: deliver prayer 20-min alerts as time-sensitive notifications, and deliver the 6:00 AM morning routine push with a deep link to the Morning Routine screen. No App Store submission required — install via Xcode direct device install.